

Module 1: Introduction to Artificial Intelligence

- What are the various types of AI?
- List out foundation of AI and describe any two in detail.
- Illustrate application areas of AI. (*Note: This question appears twice in the document*)
- Explain the components of the AI Program.

Module 2: Intelligent Agents

- Explain the PEAS description for Automated taxi driver.
- Write a short note on Goal based agent with a diagram.
- Explain problem formulation steps with water jug problem in which a full 4-gallon jug and an empty 3-gallon jug, the goal is to fill the 4 gallon jug with exactly 2- gallons of water.
- What is a PEAS description. Give peas description for a robot meant for cleaning the house?
- Explain all the AI agent with diagram.
- Explain various properties of the task environment with suitable examples.
- Describe in detail a problem solving agent with a diagram. What is the difference between Problem solving agent and Planning agent?
- Explain the different types of environments of AI agents.
- Describe PEAS description. Illustrate the PEAS description for Vacuum Cleaner Agent.

Module 3: Problem Solving

- Give the difference between informed search and uninformed search algorithms.
- What is an adversarial search algorithm and a game playing algorithm?
- Define pruning? Explain alpha and beta pruning algorithm with an example.
- Write a short note on Breadth First Search.
- Elaborate Heuristic functions of 8 tile puzzle problem.
- To demonstrate minimax with alpha-beta pruning, we use the following data to generate tree. (((3 17) (2 12)) ((15) (25 0))) (((2 5) (3)) ((2 14))))
- Explain Hill climbing algorithm and problems that occur in hill climbing algorithm?
- Explain Genetic Algorithms in details. explain it with an example.
- Discuss A* Algorithm with an example.
- Apply Alpha-Beta Pruning on the following example given in Figure considering the first node as max.
- Consider the graph given in Figure below. Assume that the initial state is S and the goal state is G. Show how A* Search would create a search tree to find a path from the initial state to the goal state.
- Explain the concept of hill climbing algorithm & simulated annealing algorithm.

Module 4: Knowledge and Reasoning

- Describe levels of knowledge based agents.
- Define uncertainty in AI. What are the sources of uncertainty in real-world problems?
- Classify the types of chaining.
- Illustrate forward chaining and backward chaining algorithms with suitable examples.
- Define Belief Network. Describe the steps of constructing a belief network with an example. What types of inferences can be drawn from that?
- Distinguish between Propositional logic (PL) and first order predicate logic(FOPL) knowledge representation mechanisms. Take suitable examples for each point of differentiation.
- Describe Bayesian Belief Network in detail.
- Differentiate between first-order logic and propositional logic.

Module 5: Planning and Learning

- Differentiate between supervised and unsupervised Learning.
- Describe different forms of learning.
- What are different types of Planning? Explain any two types with an example?
- What is reinforcement learning? Explain its types.
- Explain Partial order planning with suitable examples.
- Write short note on PAC Learning.
- Write a short note on Statistical learning.

Module 6: AI Applications

- Write a short note on NLP.
- What are Problems Robots can solve.
- Explain different applications of AI in healthcare, retail and banking.